Biochar in Stormwater Best Management Practices

Technical Basis Report



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CENTER FOR WATERSHED PROTECTION

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American Biochar Institute

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PURPOSE OF THIS REPORT

Biochar, a durable, carbon-rich material, has been explored for its potential to be incorporated into stormwater best management practices (BMPs) to improve the effectiveness and resilience of those practices. There is abundant research documenting biochar performance for pollutant reduction and soil improvement in stormwater BMPs, but varying conditions, biochar type, and experimental design have made developing overarching conclusions about biochar performance in BMPs difficult. This technical basis report aims to address this gap by summarizing general biochar performance expectations in stormwater BMPs related to specific biochar characteristics. Expectations for biochar performance in stormwater BMPs are based on a literature review and interviews with stormwater and biochar experts.

This report begins with *Section 1 - Introduction to Biochar* which summarizes what is included under the term "biochar" and how biochar production conditions and feedstock impact the physical and chemical properties of the resulting biochar. In interviews with experts familiar with biochar, the material variability between biochar suppliers and material consistency in biochar sourced from individual suppliers were discussed as some of the largest obstacles for standardizing the incorporation of biochar into stormwater BMPs. This section also reviews existing product standards in the biochar sector.

Section 2 – Biochar Characteristics, Effects, and Stormwater Goals provides a review of studies that identify biochar characteristics relative to performance within BMPs, specifically related to common stormwater goals such as runoff retention, infiltration, and pollutant reduction. This section also identifies some key remaining data gaps.

Section 3 – Biochar Performance in Stormwater BMPs provides a review of the effectiveness of biochar-amended stormwater BMPs. In many cases, these studies do not investigate specific biochar characteristics but focus more on measuring stormwater performance goals using a specific biochar in a specific stormwater BMP. Section 3 provides insight into the real-world performance of biochar under typical stormwater BMP conditions.

Section 4 – Concerns, Considerations, and Knowledge Gaps Around Biochar discusses additional knowledge gaps and other concerns that are either insufficiently addressed or only indirectly referenced in existing literature. Because the use of biochar in stormwater BMPs remains relatively novel, these knowledge gaps and concerns are included as cautionary considerations rather than as limitations on the current use of biochar in stormwater BMPs. Section 2 and Section 3 provide support for biochar use in stormwater BMPs for certain applications. Section 4 is meant to highlight where biochar use may not be applicable, or where additional design considerations should be made.

This report is written in conjunction with the *Biochar in Stormwater BMPs Guidebook for Stormwater Practitioners* (LINK), developed with funding from the United States Endowment for Forestry and Communities. The guidebook provides recommendations for incorporating biochar into stormwater BMPs, including material selection, testing, and installation, and this document provides research and data supporting the guidance.

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ACRONYMS & ABBREVIATIONS

Acronym/Abbreviation	Definition
AC	Activated Carbon
ABI	American Biochar Institute
AEC	Anion Exchange Capacity
Al	Aluminum
ANSI	American National Standards Institute
BMP(s)	Best Management Practice(s)
BSM	Bioretention Soil Media
C	Carbon
Ca	Calcium
CEC	Cation Exchange Capacity
CSI	Carbon Standards International
DOM	Dissolved Organic Matter
Fe	Iron
H	Hydrogen
HSG	Hydraulic Soil Group
IBI	International Biochar Initiative
ISO	International Standards Organization
K	Potassium
K_{SAT}	Saturated Hydraulic Conductivity
MIP	Mercury Intrusion Porosimetry
N	Nitrogen
Na	Sodium
Mg	Magnesium
O	Oxygen
O&M	Operations & Maintenance
OPFRs	Organophosphate Flame Retardants
P	Phosphorus
PAC	Polyaromatic Carbon
PAH	Polycyclic Aromatic Hydrocarbon

PAW	Plant Available Water
PCB(s)	Polychlorinated Biphenyl(s)
PFAS	Per- and Polyfluoroalkyl Substances
PSD	Particle Size Distribution
S	Sulfur
Si	Silica
SSA	Specific Surface Area
TN	Total Nitrogen
TP	Total Phosphorus
TrOC(s)	Trace Organic Compound(s)
VM	Volatile Matter
WBC	World Biochar Certificate
WHC	Water Holding Capacity

SECTION 1 - INTRODUCTION TO BIOCHAR

Biochar is a black, carbon-rich material produced when biomass is heated above 350 °C in oxygen-free or oxygen-limited conditions, most commonly in pyrolysis or gasification systems. Under such conditions, a portion of the biomass is converted into combustible vapors that are released from the biomass ("pyrolysis gasses"), but because there is little or no oxygen, these pyrolysis gasses cannot combust. The remainder of the material is converted into biochar, a carbon enriched material that is essentially charcoal.

The conversion of biomass to biochar enriches the resulting material in carbon, dramatically increases the surface area and porosity of the material, and modifies the structure of carbon atoms such that they are highly resistant to microbial decay in the environment. Generally, each of these effects increases proportionally with biochar production temperature, which commonly ranges from approximately 350 °C to 900 °C.

Biochar feedstock, pre-processing, production conditions, and post-processing will each impact the physical and chemical properties of the resulting material.

Biochar Production

Pyrolysis and gasification are the most common techniques to produce biochar, but there are a range of different processes and technologies that effectively produce biochar Mohanty et al., 2018, Pencha et al., 2021). Each production technology produces biochar of different properties. Common commercial production technologies for biochar include slow pyrolysis, fast pyrolysis, gasification, combustion (in biomass energy boilers), and air curtain carbonizers (IBI and USBI, 2024, Mašek et al., 2024). A summary of key attributes of these technologies is presented in Table 1.

Biochar Physical and Chemical Properties

When biomass (i.e. wood chips, crop residues) is heated above 300 °C in the absence of oxygen (pyrolysis), its carbon atoms reorganize into hexagonal "aromatic" rings that fuse into polyaromatic carbon (PAC) structures (Keiluweit et al, 2010). These PAC structures are highly resistant to biological and chemical breakdown, so a large fraction of biochar carbon can remain in soils and sediments for hundreds to thousands of years (Woolf et al., 2021). The carbon (C) within these PAC structures is referred to as "fixed carbon" (fixed C) due to its stability in the environment.

Table 1. Summary of biochar production processes

Technology	Oxygen Availability	Peak Temperature (°C)	Residence / Hold Time	Pyrolysis Gas Fate	Feedstock Flexibility	Char Mass Yield (%, dry basis)	Carbon Yield (% of feed C)
Slow Pyrolysis	Low or no oxygen	~400–650	10–60 minutes	Combusted in adjacent chamber for heat and/or electricity production	Broad range of particle size, morphology, moisture,	~15–30	~30-55
Fast Pyrolysis	Low or no oxygen	~450–550	less than 30 seconds	Condensed into bio-oils	Fine, uniform particles <2–6 mm; ≤10–12% moisture	~10–20	~20–35
Gasification	Intentionally metered oxygen insufficient for full combustion	~750–950	5-30 minutes	Portion combusted in production chamber to heat process and portion combusted in adjacent chamber for heat or electricity	Dry (≤15–20%), size-controlled;	~5–20	~10–35
Combustion (Boilers)	Excess, but feedstock variability prevents 100% combustion	~800–1,100+	Seconds–minutes	Combustion of vapors and biochar for heat production is primary goal	High consistency improves combustion and heat production efficiency	<5	<10
Air Curtain Carbonizers	Excess air with air curtain combusting vapors above char	~900–1,100	Minutes–hours (batch)	Combusted above the biochar in the air curtain, producing heat for pyrolysis	Coarse/heterogeneous woody debris; minimal preprocessing	~5-15	~10-25

Main Components of Biochar

Biochar is typically described in terms of three main fractions: fixed C, ash, and volatile matter (VM). The composition and size of each fraction is determined by production temperature and biochar feedstock. Variance in each fraction influences how biochar behaves in stormwater systems.

Fixed Carbon

Fixed C is the stable PAC matrix formed during pyrolysis. Its proportion increases as production temperature increases (Keiluweit et al, 2010). This stable carbon is a key reason biochar is long-lived and useful for long-term carbon storage.

Ash

The ash fraction is determined by what remains after complete combustion of biochar. As production temperature increases, more of the original biomass is volatilized or combusted, and the proportion of ash in the remaining biochar increases (Domingues et al., 2017).

The ash content depends on the feedstock but commonly contains:

- Oxides, hydroxides, carbonates, and phosphates of calcium (Ca), magnesium (Mg), potassium (K), and sodium (Na),
- Silicates and aluminosilicates, and/or
- Aluminum (Al) and iron oxides (Vassilev et al., 2013).

Hydroxides and carbonates in the ash provide alkalinity and tend to keep biochar pH above neutral. When biochar is used in stormwater BMPs, this liming effect can be useful (or in some cases problematic). See Section on pH and Liming for examples of liming impacts on stormwater BMPs.

Volatile Matter

Volatile matter (VM) is estimated as the portion of biochar that is removed via volatilization when the material is heated to~900 °C in a standardized lab test. As production temperature increases, VM content generally decreases due to volatilization.

In biochar, volatile matter includes biomass tars and oils, as well as various compounds containing oxygen (O) and hydrogen (H) (Keiluweit et al., 2010). Additionally, most of the nitrogen (N) originally present in the biomass is removed via volatilization (Keiluweit et al., 2010). Higher VM content typically indicates a less carbonized biochar with more residual organic compounds.

Elemental Composition and Stability Indicators

Biochar is made up mainly of carbon (C), oxygen (O), hydrogen (H), and nitrogen (N), along with mineral elements such as calcium (Ca), sulfur (S), magnesium (Mg), silica (Si), phosphorus (P), potassium (K), iron (Fe), aluminum (Al), and sodium (Na) that are present in the ash fraction (Keiluweit et al., 2010). As production temperature increases:

- C content and ash-associated elements (e.g., Ca, Mg, K, P) tend to increase.
- O, H, and N contents tend to decrease as they are volatilized (Keiluweit et al., 2010).

Although the total content of ash-associated elements is important to know, their fate in the environment depends on the element's chemical form. For example, some N in biochar is present as soluble nitrate, nitrite, and ammonium, but the majority is typically contained in organic N forms which can be slowly released

through microbial degradation (Clough et al., 2013), but this process is much slower than for nitrogen contained in typical biomass because biochar only degrades very slowly. Whereas P in biochar is commonly contained in relatively low solubility minerals in the ash fraction. In low-ash biochar, the minerals dissolve very slowly and at low concentrations (Ippolito et al., 2020), such that the overall water quality risk from phosphorus is typically quite low (Novak et al., 2018). Leaching tests can be used to better understand the potential fate of ash-associated elements. However, it is recommended that testing modifications such as using a neutral leaching solution or pre-washing biochar material be incorporated to avoid conclusions related to the initial washing of material.

Higher pyrolysis production temperatures result in an increased abundance of recalcitrant PAC structures but lower concentrations of O and H. The H:C ratio[1] is the most commonly used proxy for biochar stability (Azzi et al., 2024, Woolf et al., 2021), with a maximum threshold ratio of 0.7. An H:C ratio of ≤ 0.7 identifies a material as biochar and signifies the material's very resistance to microbial decay in the environment (Ippolito et al., 2020).

The H:C and H:O ratios are useful screening tools when selecting biochar for long-term applications, such as stormwater infrastructure, where media longevity and carbon persistence are important considerations.

Physical Structure and Porosity

The physical appearance of biochar largely reflects the original feedstock. Biochar often resembles a slightly shrunken, darker version of the initial biomass material (Sun et al., 2026). This is due to the loss of mass during volatilization and condensation of carbon into a more compact PAC structure during high-heat production.

At finer scales, biochar retains many of the structural features of the original biomass. For plant-derived biochar, the cellular structure is often preserved. Wood-based biochar, for example, retain a well-ordered vascular structure, creating a highly porous material with **macropores** typically in the range of 10–40 µm. Macropores are good for increasing infiltration pathways within soil and provide habitat for biofilms. At the nanoscale, most biochar can develop abundant **micropores**, many as small as 2 nm. These micropores are especially common in biochar produced at temperatures above ~600 °C and are responsible for most of the very high specific surface area (SSA) observed in high temperature biochar (up to ~400 m^2/g) (Gray et al., 2014). Micropores provide essential surface area for sorption of pollutants. For reference, a well-accepted and effective but expensive treatment media for pollutant capture and removal - activated carbon (AC) – typically has SSA in the range of 1000 to 3,500 m^2/g (Chen et al., 2008, Li et al., 2023). The combination of macropores and micropores is particularly important for stormwater applications, where both hydraulic performance and pollutant removal are critical.

Effect of Production Temperature (Summary)

The chemical and physical properties of biochar are strongly influenced by production temperature, which is one of the most important levers in designing biochar for specific stormwater uses. In general, as production temperature increases, the properties listed in Table 2 tend to increase or decrease as shown.

[1] H:C and O:C ratios are reported as atomic ratios, not weight ratios. For example, an H:C ratio of 0.4 indicates 4 hydrogen atoms for each 10 carbon atoms.

Table 2. Summary of the impacts of temperature on the chemical and physical properties of biochar.

Properties that **Increase** with Increasing Production Temperatures	Properties that **Decrease** with Increasing Production Temperature
C and fixed C content	VM content
Ash content	O, H, N content
pH and liming capacity	H:C ratio; O:C ratio
Microporosity and SSA	Surface functional group density
Resistance to microbial degradation	Initial biochar cation exchange capacity (CEC)

Biochar Feedstock

The biomass used to produce biochar is commonly known as feedstock and can include waste materials such as crop residues, agricultural processing wastes, timber processing residuals, forest residues, urban yard and forestry waste, food waste, animal manures, and wastewater treatment plant biosolids. To simplify, feedstock can be categorized as woody/wood-based, herbaceous, or manures/biosolids. The type of feedstock influences many biochar properties as simplified in Table 3.

Wood-based feedstocks have high lignin content, which provides structure to biochar, resulting in biochar with higher SSA than other feedstocks. Grass and biosolid feedstocks result in biochar with high ash and mineral content, which can increase biochar reactivity with cations in soil or water through chemisorption.

Table 3. Summary of feedstock type and associated biochar physical and chemical characteristics.

	Woody	**Herbaceous**	**Manures / Biosolids**
Example Feedstocks	Sawdust Wood chips Nut shells	Rice hulls Corn stover Wheat straw	Chicken litter Dairy manure Biosolids
Carbon content	High	Low – Moderate	Low – Moderate
Ash content	Low	Moderate – High	Moderate – High
pH	Mildly alkaline	Alkaline	Highly alkaline
Electrical conductivity	Low	Moderate	High
Surface area potential	High	Moderate	Low – Moderate
Microporosity	High	Moderate	Low – Moderate
Physical hardness	High	Low – Moderate	Low
Nutrient content	Low	Low – Moderate	Moderate – High
Bulk density	Low	Low – Moderate	Moderate – High

However, biochar of any feedstock that are produced at lower temperatures may contain undesirable compounds, such as nutrients and polycyclic aromatic hydrocarbons (PAHs), which may leach into water and

soils. At any pyrolysis temperature, plant-based biochar feedstocks tend to be cleaner than sludge or manure-based biochar, which has the potential to leach minerals in stormwater applications (Mohanty et al, 2018, Novotný et al., 2023). Of the plant-based biochar, wood-based biochar tends to have the least nutrients available for plant growth but also lower nutrient leaching potential (Novotný et al., 2023). Therefore, unless otherwise advised by experts in stormwater management and biochar, it is recommended that stormwater applications use biochar produced at high temperatures from high carbon feedstocks (typically wood-based) due to their low potential for leaching nutrients and other pollutants (Mohanty et al, 2018).

Biochar Regulation and Characterization

The biochar industry has developed primarily over the past two decades, and as a result, relatively few industry regulations have been established in the United States. To fill this need, the International Biochar Initiative (IBI) developed a Biochar Certification Program as a voluntary program for standardizing biochar characterization and products. IBI initially developed the *Standardized Product Definition and Product Testing Guidelines for Biochar that is Used in Soil* in 2012, and updated these standards in 2015 (IBI, 2015). This standard provides best management guidelines for biochar feedstock, production, storage, and use. Additionally, the document provides testing guidance, which is ultimately useful for classifying biochar for different end-uses. Certification requirements include annual testing of biochar properties, including moisture, carbon content, H:C, total ash, total nitrogen, pH, electrical conductivity, and particle size distribution (PSD).Additionally, the IBI Biochar Certification Program requires a tri-annual toxicity test, which includes testing for the leachability of a variety of potential environmental pollutants. Retesting is required if the feedstock changes, which is a common occurrence especially amongst smaller-scale biochar producers.

As of 2024, the IBI Certification has been discontinued, and efforts to standardize biochar materials and production have shifted to the World Biochar Certificate (WBC), American Biochar Institute (ABI), and Carbon Standards International (CSI).

WBC guidelines were developed based on the existing European Biochar Certificate (Schmidt et al., 2024). Biochar is classified as either admissible or inadmissible by intended use, based on applicable safety regulations and relevant industry standards. Parameters required for testing under the basic biochar classification for this certification include composition, pH, SSA, water holding capacity (WHC), electrical conductivity, bulk density, and water/dry matter content. Additional testing is required depending on the intended biochar application. While WBC standards and certification can be valuable as a guideline, these programs are built on European laboratory standards and European environmental regulations, making them impractical to use in North America. (Interview: Hegberg).

Starting in 2023, ABI has been developing biochar analysis standards and certifications for biochar in North America, with a focus on developing standards that can be used by North American laboratories, that comply with North American regulations, and that are developed through formal Standards Development Organizations (e.g., American National Standards Institute; ANSI). As of November 2025, ABI recommends that biochar producers and end-users ensure that samples are analyzed according to *USBI / ANSI S668: Methods for the Analysis and Testing of Biochar* (LINK). This ANSI standard recommends testing methods for common biochar properties that can easily be performed by accredited laboratories in North America.

Biochar Sourcing

The reliable availability and product consistency of biochar have historically been significant obstacles to increasing the use of biochar in stormwater management and other applications. However, in the past five to ten years, the increase in large biochar producers that are able to achieve high levels of consistency and high production volumes have reduced these barriers. The growth of these larger producers has been driven by the development of new markets for biochar, such as stormwater management and carbon financing.

Biochar carbon removal credits have recently been made available to the carbon market (IBI & USBI, 2024). These carbon removal credits are now a common feature of biochar producer business models, allowing companies to subsidize the cost of biochar, leading to lower prices. With the increasing number of biochar producers, identifying appropriate and effective biochar can be a challenge. To support the sourcing of quality biochar materials, ABI maintains a registry of biochar producers on its website at https://biochar.org/.

Section 2 – Biochar Characteristics, Effects, and Stormwater Goals

Biochar has been a product of interest in the stormwater field since 2014, when it was introduced into stormwater pilot projects for reducing heavy metals in the Pacific Northwest (Gray et al., 2015) and concomitantly on nutrient reduction initiatives within the Chesapeake Bay watershed[2] Biochar has historically been used as a soil amendment in agriculture, and a large amount of research is targeted for this application. Although not always directly related to stormwater, research on biochar in different applications does provide insight into how its different characteristics can impact how biochar interacts with water and pollutants, and how it might impact vegetation in stormwater BMPs. This section synthesizes studies to identify these characteristics and effects in relation to common stormwater goals. Results of this synthesis are summarized in Table 4 on the following page.

Cation and Anion Exchange Capacity

Biochar frequently has greater cation exchange capacity (CEC) and porosity than in-situ soils, which encourages heavy metal and organic contaminant (i.e. PAHs) adsorption, but they tend to have only limited anion exchange capacity (AEC) in normal soil (Lawrinenko and Laird, 2015). Among common stormwater pollutants, heavy metals are strongly sorbed by CEC in biochar, soil, and other components of bioretention soil media (BSM) like compost, whereas pollutants like nitrate and phosphate are strongly sorbed by anion exchange capacity.

CEC in biochar is dependent on the presence and availability of oxygen-containing functional groups on biochar after pyrolysis, as well as those subsequently formed from biotic and abiotic oxidation in soils. The CEC associated with such functional groups depends on the protonation status of the groups, such that at lower pH those sites are occupied by protons (i.e., H^+ ions) whereas at higher pH they are not. This effect results in pH-dependent CEC in biochar, with pH values above ~6.0 corresponding to higher CEC (Fidel et al, 2018). Because biochar pH is almost always alkaline, it typically increases the pH of soils and bioretention media

[2] https://inlandbays.org/projects_issues/anchorage-canal-drainage-area-stormwater-retrofit/

above neutral, ensuring that biochar provides significant CEC in most soils and in stormwater management BMPs.

Table 4. Summary of the impact of wood-based, high heat biochar characteristics on stormwater goals. A description of the characteristic measure is provided for the associated stormwater goal. Note: parentheses "( )" denote is inferred data.

Stormwater Goal	Biochar Characteristic					
	CEC	Porosity	Hydro-phobicity	Particle Size	Specific Surface Area	Age[b]
Heavy Metal Removal	High	(Moderate)	(Low)	Moderate	(Moderate)	Aged
Nitrogen Removal	High	High	(Low)	(Low)	(Moderate)	Fresh
Phosphorus Removal	(Low)	(Low)	(Low)	(Low)	(Low)	N/A
Bacteria Removal	(Moderate)	(High)	High	(Moderate)	High	Fresh
Emerging Contaminant Removal	(Moderate[a])	High	High	Moderate	High	(Moderate[a,c])
Infiltration	N/A	High	N/A	High	Low	N/A
Water Retention	N/A	Moderate	Low	N/A	Low	N/A
Plant Survivability	(Moderate)	High	Low	(Moderate)	High	(Fresh)

[a] Impact is dependent on specific emerging contaminant.

[b] Aging can be achieved prior to application via composting or after application over time in soil.

[c] Non-polar contaminant removal – fresh biochar, polar contaminant removal – aged biochar.

AEC in biochar is typically low. Like CEC, AEC is pH-dependent. However, pH values above ~5 correspond to lower AEC and pH values below ~5 result in higher AEC. As noted, because soils with biochar typically have a pH at or above neutral (i.e., 7.0), AEC from biochar is typically very low (Lawrinenko and Laird, 2015).

While high-mineral biochar tends to have a larger variety of functional groups, they are not recommended for general stormwater applications due to potential leaching (Novotný et al, 2023). The quantity of functional groups in herbaceous and woody biochar is most correlated to pyrolysis temperature and ageing, where lower temperature biochar tends to have higher CEC immediately after production (Lu et al., 2023, Mukherjee et al., 2011, He et al., 2024). Alam and Anwar (2020) found that eucalyptus wood waste biochar – produced via pyrolysis at 400°C for 10 min and then both wet and dry sieved at 2.36 mm to reduce fines – reduced NH_4^+ (cation) by nearly 99% in a flask adsorption test. However, there was little to no reduction in NO_3^- or PO_4^- (anions). This indicates that adsorption strongly preferred cations due to available functional groups at the pH of the biochar in water. Fassman-Beck (2025) found a direct correlation between amendment CEC, where amendments included regenerated activated carbon and biochar, and Cu removal in stormwater. This suggests that biochar with high CEC can remove positively charged pollutants.

Over time, surface functional groups form on all biochar due to the biotic and abiotic oxidation of biochar surfaces. It is possible that over longer time frames, biochar produced at higher temperatures and containing greater porosity and surface area could develop the greatest CEC. Therefore, additional research is necessary

to understand the overall performance of biochar in stormwater management applications based on the initial biochar CEC.

Porosity

Porosity is critical for a range of effects in stormwater systems, including water retention, microbial habitat, and pollutant removal (Erickson et al., 2016). Porosity in biochar can be broken into three distinct categories (Gray et al., 2014) (Figure 1):

- External pores (Inter-porosity): Pores between biochar particles or between biochar particles and other soil or media components (e.g., sand, soil, compost).
- Residual macropores (Intra-porosity): Internal pores that retain the size and shape of feedstock structures.
- Pyrogenic nanopores: Internal pores that are formed at higher production temperatures and are typically in the nanometer size range.

External pores are determined by the size and shape of biochar and other components of the soil or engineered media. Residual macropores reflect the size and shape of the structure of the feedstock, albeit at a smaller size due to the physical shrinkage of biochar during production. Accordingly, different feedstocks have different potential for residual macroporosity, which represents most of the overall internal pore volume of biochar particles (Sun et al., 2012). Macropores tend to range in size from 1 to 100 micrometer in diameter which are ecologically and hydrologically relevant. Pyrogenic nanopores are formed at high production temperatures due to transformation of biomass into smaller forms, leaving behind nanometer sized voids. These pyrogenic nanopores represent only a small portion of the overall internal porosity of biochar but represent the majority of biochar surface area and nutrient and pollutant sorption sites (Li et al., 2008), for smaller compounds that can access these surfaces.

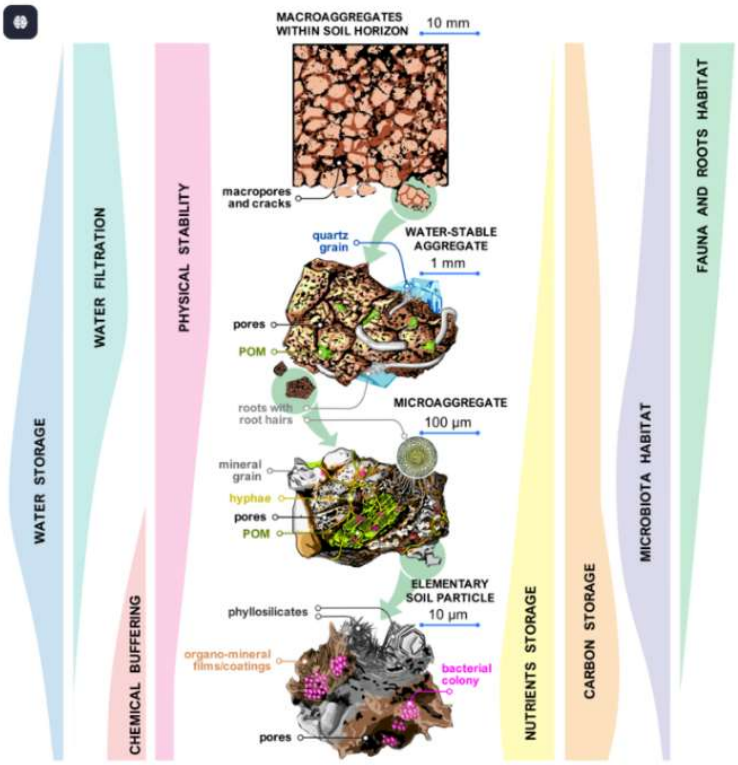


Figure 1. Graphical depiction of intra-, inter-, and nanopores within soil (Yudina and Kuzyakov, 2023).

Specific Surface Area

Specific surface area (SSA) describes the amount of surface area available for physical, biological, or chemical interactions on a biochar particle, which can be beneficial for pollutant removal (He et al., 2024) and water retention (Chen et al., 2021). Higher biochar production temperatures lead to the reorganization of carbon into more physically compact polyaromatic forms, which, in turn, creates voids as pyrogenic nanopores, leading to increased specific surface area (Lu et al., 2023). Because carbon becomes increasingly more polyaromatic with increasing production temperature, specific surface area typically increases with increasing production temperature (Keiluweit et al, 2010).

Biochar can have specific surface area values as high as 400 m^2/g or more (Keiluweit et al., 2010). However, compared to activated carbon, which is among the most adsorbent filtration media components, biochar has significantly lower surface area, but is often much cheaper and more appropriate in stormwater BMPs.

SSA and porosity are often used interchangeably, but they impact biochar performance differently. Porosity is a measure of the volume of pore space, which includes macro- through nano-pores. Porosity is important for improving water holding capacity and infiltration within soil or media. SSA, on the other hand, measures pore wall area for reactions and adsorption necessary for pollutant reduction and nutrient retention. Large pores will increase porosity, whereas small pores will increase SSA.

Hydrophobicity

The level of hydrophobicity of a biochar material describes the material's affinity to water. High hydrophobicity material has less affinity for water than a low hydrophobicity (or hydrophilic) material. In porous materials, hydrophobicity can limit the ability of water to enter pores due to repellent capillary forces (Gray et al., 2014). Accordingly, hydrophobicity impacts what compounds and molecules have access to biochar surface area and sorption sites, particularly those sites in residual macropores and pyrogenic nanopores. Hydrophobic pollutants, such as biological contaminants and some trace organic compounds (TrOCs) are themselves hydrophobic and will undergo hydrophobic attachment to biochar surfaces (Afrooz and Boehm, 2017, Chen et al., 2008, Ulrich et al., 2017). So, while hydrophobic biochar permits less water to enter its pores, it can also be more effective at removing certain pollutants that *do* enter those pores. On the other hand, less hydrophobic biochar tends to more easily allow and hold water within the micropore spaces, especially pyrogenic nanopores – which contain the majority of surface area in biochar (Khan et al., 2023). In general, less hydrophobic biochar is typically more effective for general stormwater treatment applications.

Hydrophobicity in biochar is related to the presence of residual tars and oils and PAC structures, which have low affinity for water. Hydrophobicity from tars and oils is more common in biochar produced below ~ 500 °C (Gray et al., 2014). This is, especially, the case for biochar produced in systems with poor vapor removal which can allow the recondensation of vaporized tars and oils onto biochar when they cool. Hydrophobicity associated with the low water affinity of PAC structures increases with increasing production temperatures due to the declining presence of oxygen-containing hydrophilic functional groups (Adhikari et al., 2022, Lu et al., 2023). Because the presence of oxygen-containing functional groups reduces hydrophobicity, high O:C ratios can be an indicator of hydrophilic properties, as they reflect a higher proportion of oxygen-containing functional groups relative to the carbon structure (Lu et al., 2023).

Both of these factors are likely to decrease with ageing. Adhikari et al. (2022) showed that biochar age does in fact reduce biochar hydrophobicity even over the course of a single year. Specifically, the presence of tars and oils on biochar surfaces decrease over time as they are slowly dissolved and consumed by soil microbes. Similarly, as noted, the microbial oxidation of biochar surfaces slowly increases the abundance of oxygen-containing functional groups. A biochar that has been aged by one year has reduced hydrophobicity compared to its fresher counterparts.

Particle Size Distribution

Biochar particle size is important to consider for stormwater quality and quantity goals. Smaller particles have more desirable pollutant removal properties. These properties include increased external surface area and porosity (Boehm et al., 2020, Chen et al., 2022), and better connectivity between pollutants in water and sorption sites in internal pores. However, small biochar particles can reduce infiltration rate, which negatively impacts runoff quantity management, as well as pollutant load reduction dependent on volume reduction. This is also a common concern for other stormwater filtration media and a key reason why many stormwater media specifications limit the presence of fine particles (Kaya et al., 2022).

The small external pore sizes of smaller biochar may also improve water retention by modifying soil texture (Edeh et al., 2020, Adhikari et al., 2022). However, as noted, smaller biochar particles are not effective for improving stormwater infiltration rates as they have the potential to clog sand-based media systems and BMPs that filter water at higher rates (Liu et al., 2016, Le et al., 2020, Nakhli and Imhoff, 2021). It is important to note that in cases of sandy medium, decreasing infiltration rates may be desirable for increasing retention and the associated pollutant removal. However, in the case of typical stormwater media (which tend to be sandy), biochar particle sizes should be similar to the overall media particle size distribution (PSD) to prevent migration and clogging. For stormwater BMPs that filter water (i.e., bioretention with underdrains, green roofs), discharge back to surface waters, and/or those that infiltrate water into sandy soils, the biochar PSD should generally be between 0.5 mm and 2 mm in size, with only a slight percentage of the total composition consisting of biochar smaller than 0.075 mm (Nakhli and Imhoff, 2021). For stormwater BMPs that promote general water storage or infiltration into underlying soils that have finer textures, the particle size distribution of biochar is less critical.

pH and Liming

Biochar pH and liming are key material properties in agricultural applications. However, they typically have only a limited and short-term effect on the performance of stormwater BMPs when used in blended media. Due to the presence of ash compounds, the majority of biochar is alkaline, particularly those made at higher production temperatures and from high-mineral feedstocks (Ippolito et al., 2020). These ash compounds – frequently composed of carbonates, oxides, and hydroxides (Vassilev et al., 2013) – slowly dissolve in water and remove protons from the solution. Therefore, the solution pH increases. This alkaline pH can increase the CEC of biochar and other components with pH-dependent CEC (e.g., compost), leading to improved removal of heavy metals and other cationic pollutants (Li et al., 2017). Further, at aqueous pH values above ~ 8.5, some metals can be removed from water via precipitation reactions as hydroxides (Benalia et al., 2022).

While the pH of most biochar is alkaline and can reach above 10, most biochar has a liming capacity of less than 10% calcium carbonate equivalence ($CaCO_3$-eq) (Adewumi et al., 2025). So, it will only moderately increase the pH of a blended stormwater bioretention media or amended native topsoil. Further, most

stormwater BMPs are designed with relatively large impervious catchments resulting in high hydraulic loading ratios, meaning that any ash-related pH effects from biochar will tend to be short-lived as these compounds slowly dissolve into runoff passing through the BMP. Accordingly, in practice, the pH of stormwater BMPs tends to trend towards the pH of the influent stormwater (International Stormwater BMP Database, LINK).

Biochar Ageing and Inoculation

Raw or fresh biochar typically exhibits relatively low CEC compared to aged biochar, particularly fresh biochar produced at higher production temperatures where many oxygen-containing functional groups are volatilized. Conversely, raw, high-temperature biochar does typically exhibit a high affinity for hydrophobic compounds. However, over time, with slow biotic and abiotic ageing, biochar develops increased surface functionality, and particularly negatively charged oxygen-containing functional groups that increase biochar CEC. This effect has been documented in laboratory incubation experiments (Cheng et al., 2006) and by sampling biochar collected from soil along a climosequence (or soil horizons varying due to differences in climactic conditions) of traditional charcoal production sites (Cheng et al., 2008).

The ageing of biochar can be accelerated, most notably by adding biochar to other waste biomass during composting. When biochar is "co-composted" in this way, the addition of biochar can accelerate the composting process and reduce odor and gaseous pollutants production. The biochar itself rapidly becomes aged, likely due to the high heat and microbial activity in an active compost pile, resulting in biochar with increased CEC and surface functional group density (Godlewska et al., 2017). While this approach can be effective, co-composted biochar may not be appropriate for some stormwater applications because compost has been shown to leach significant amounts of nutrients which may be harmful for surface water and groundwater. Co-composting could be an option for applications without direct connections to storm drain systems, surface or ground waters, such as soil amendment or bioretention without an underdrain.

Biochar Impacts on Physical Filtering and Sedimentation

Many stormwater pollutants are present in particulate forms as opposed to dissolved forms. This is particularly true for nutrients and trace organic (TrOCs) pollutants. Nutrients, both nitrogen and phosphorus, are inherent components of organic matter and are frequently present in stormwater as particulates of organic matter. Trace organics, such as PCBs and PAHs, are frequently sorbed to organic matter. Removal of these particulates – which does not include dissolved nutrients – in stormwater systems is most commonly achieved via sedimentation and physical filtration.

Biochar can be used in stormwater BMPs that achieve sedimentation, but it has no direct effect on the process itself. Biochar is also commonly used in BMPs that remove pollutants via physical filtration. While biochar is an effective media for this process, it is not likely more effective than other inert filtration media such as sand.

SECTION 3 - BIOCHAR PERFORMANCE IN STORMWATER BMPS

Nationally, stormwater goals vary by region but generally relate to flow mitigation and pollutant reduction. Maintaining vegetation health can also be important in terms of beautification, maintaining public support for stormwater green infrastructure, and managing operations and maintenance (O&M) costs.

Common pollutants of concern may include heavy metals, nitrogen, phosphorus, bacteria, and trace organic compounds (TrOCs). Below is a summary of biochar's primary mechanisms for removal by the type of pollutant. More details are provided after Table 5.

- ***Heavy metals*** are removed from the water column through adsorption via cation exchange or surface complexation via functional groups (Guo et al., 2020).
- ***Nitrogen*** is found in different phases within stormwater – Nitrate/nitrite, ammonia/ammonium, and organic nitrogen. Ammonium can be removed via adsorption to negatively charged functional groups comprising biochar CEC. Nitrates/nitrites can be removed via anion exchange, but as noted, anion exchange capacity in biochar (and in other soil components) is typically low. In practice, nitrate and nitrite can be removed via denitrification in micro-anoxic zones within biochar pores where this process can occur. Organic nitrogen can be removed via physical filtering, which is relatively unimpacted by the addition of biochar. Organic nitrogen can also be removed via plant uptake and degradation through microbial activity supported by healthy plants and root zones, both of which could be enhanced by biochar. However, the overall increase in nitrogen removal is likely to be small.
- ***Phosphorus*** can be found in stormwater in particulate form or as phosphate. Phosphate can be sorbed by positive functional groups comprising anion exchange capacity, but this tends to be limited in biochar. Particulate phosphorus is typically removed via physical filtering, which is relatively unimpacted by the addition of biochar. Organic phosphorus forms can be made available for plant uptake through microbial breakdown. Overall, the impact of biochar on phosphorus is limited, but it can enhance phosphorus removal with increased anion exchange capacity or by supporting healthy plant and root zones.
- ***Bacteria*** often includes both indicator (E. coli, Enterococci, etc.) and pathogenic organisms, which differ in size, shape, and hardiness. Generally, bacteria can be filtered or trapped in media columns, where they may ultimately suffer desiccation, predation, and/or inactivation. Bacteria are generally hydrophobic, so they tend to sorb weakly to hydrophobic surfaces. The increased porosity and surface area of biochar, enhanced by the hydrophobicity in some biochar, increases the potential for trapping bacteria within soil or engineered media. There is potential for biological contaminant growth from nutrients within a stormwater system (Chudoba et al., 2013) and their subsequent release back into the environment, though the majority of harmful bacteria trapped in filtration media are either consumed by other bacteria, or die (Chandrasena et al., 2014).
- ***Trace Organic Compounds (TrOCs)*** include a variety of contaminant categories such as pesticides, herbicides, PAHs, and per- and polyfluoroalkyl substances (PFAS). Within this broad group of contaminants, some, like PAHs and PCBs, are commonly sorbed onto organic particulate matter, while others, such as PFAS, are more commonly dissolved. Accordingly, biochar has little or no benefit (over sand) for the filtration and sedimentation removal of those substances contained in particulate matter. On the other hand, biochar can be an effective sorbent for dissolved trace organics, as they can be

adsorbed to biochar, predominantly via electrostatic attraction, hydrophobic effects, cation exchange, and hydrogen bonding (Hegberg et al., 2024). Once adsorbed, biochar may provide a supportive environment for microbial degradation of the pollutants.

In understanding how stormwater pollutants of concern are removed, there are several additional stormwater BMP goals that come to light that support pollutant removal, such as runoff reduction through infiltration or retention, WHC, and plant survivability, all of which biochar amendment can impact.

- ***Infiltration*** reduces stormwater runoff into waterbodies by slowly infiltrating it through the soil medium. This disconnects associated pollutants from a direct connection to surface water and provides more opportunity for natural removal processes to occur. Further, most pollutants, including those that are particulate in nature and those that are effectively sorbed via cation exchange, typically exhibit little or no migration to or within groundwater. Among common pollutants, nitrate has the greatest potential for migration to and within groundwater, and from there towards surface waters.
- ***Water Holding Capacity (WHC)*** is important for detaining or retaining runoff for extended periods of time. This allows for natural pollutant removal processes to occur (i.e. denitrification in micro-anoxic pore space) and supports plant communities, which in turn provide many supporting benefits to BMPs. Increased water holding capacity can also reduce the total amount of water that discharges from flow through BMPs (e.g., bioretention with underdrains, green roofs) by increasing the amount of runoff that is evapotranspired by vegetation.
- ***Plant Survivability*** is critical because plants support many pollutant removal mechanisms within media and soil, including hosting microbial communities and facilitating nutrient uptake. Plant survivability is also crucial for public support (e.g., beautification and green spaces) of stormwater green infrastructure in general and is an important driver of O&M costs of stormwater BMPs. In drought-prone areas or dense urban areas where BMPs are surrounded by impervious surfaces, plant maintenance in BMPs involves watering, mulching, mowing, and the replacement of dead vegetation. All of which can account for considerable costs for municipalities and landowners.

Table 5. Summary of expected performance by biochar-amended BMP type on stormwater goals. ND = no data; ( ) = inferred performance; N/A = goal does not apply to stormwater BMP; ↑=increase; ↓ = decrease; ↔ = studies show variable results.

Stormwater Goal	Biochar Impact on BMP Performance						
	Bioretention	**High Flow Media Filter**	**Vegetated Swales**	**Green Roof**	**Soil Amendment**	**Filter Strip & Roadside Embankment**	**Dry Pond Retrofit**
Heavy Metal Removal	↑ Cu, Ni, Pb ↑↔ Cd ↔ As, Cr, Zn	↑ Cu, Ni, Pb ↑↔ Cd ↔ As, Cr, Zn	ND	N/A	N/A	ND	ND
Nitrogen Removal	↑ TN w/ saturated zone ↑↔ TN w/out saturated zone ↑ NO3/NO2 ↓↔ TP	↑ TN w/ saturated zone ↑↔ TN w/out saturated zone ↑ NO3/NO2 ↓↔ TP	(↑ TN)	↓TN export ↓NO3 export	N/A	↑ TN	(↑ TN)
Bacteria Removal	↑ E. coli ↑ Enterococci	↑ E. coli ↑ Enterococci	ND	ND	N/A	ND	ND
TrOC Removal	↑ TrOCs	↑ TrOCs	ND	ND	N/A	ND	ND
Infiltration	↔ Ksat	↔ Ksat	↑ Ksat ↑ Volume Reduction	↓ Ksat	↑ Ksat, fine soils ↓ Ksat, coarse soils ↓ Bulk Density ↑ Volume Reduction	↑ Ksat, fine soils ↓ Ksat, coarse soils ↓ Bulk Density ↑ Volume Reduction	↑ Ksat, fine soils ↓ Ksat, coarse soils ↓ Bulk Density ↑ Volume Reduction
Water Retention	↑ WHC ↑ PAW	↑ WHC ↑ PAW		↑ WHC ↑ PAW	↑ WHC	↑ WHC	↑ WHC
Plant Survivability	↑ Plant Height ↑ Number of Shoots	↑ Plant Height ↑ Number of Shoots	↑ Plant Coverage	↑ Soil Microbes ↑ Eukaryote ↑ Plant Biomass	↑ Foliage N ↑ Plant Coverage	↑ Folier N ↑ Plant Coverage	↑ Folier N ↑ Plant Coverage

Heavy Metals

Many research studies show improved heavy metal removal in biochar-amended soils (Guo et al, 2010, Beesley et al., 2011, Khan et al., 2023, Kumar et al., 2023). However, many of these studies applied biochar for remediation or urban soil improvement, and not in stormwater-specific applications. Concentrations of heavy metals in typical urban stormwater are typically orders of magnitude lower than concentrations in pore water at contaminated sites. Adsorption curves are driven by contaminant concentration, with higher sorption at higher concentrations, so stormwater professionals should be cautious about how industrial or remediation data is used.

In a research review, Boehm et al. (2020) found only 14 papers as of September 10, 2019, that met the screening criteria for isolating the impacts of biochar amendments in media. Of those 14 studies, only between one and six data points could be extrapolated for heavy metals (Ray et al., 2019, Zhoa et al., 2019). These data showed that, in short-term trials, when using high-temperature wood-based biochar in sand media filters (without compost amendment), Cu, Ni, and Pb $\log_{10}$ removal was higher than control columns without biochar amendment (i.e. sand-only). Similar results were observed in a review by Mohanty et al. (2018) using a different set of studies with activated/amended biochar material. Pb, Cu, and Zn have a high affinity for organic complexation and are, therefore, preferentially adsorbed when organic matter (i.e. compost) or high-mineral biochar is used. This is particularly the case for ionic metals such as Zn (Guo et al., 2010, Mohanty et al., 2018, Hegberg et al., 2024, Spahr et al., 2022, Ulrich et al., 2023). While these data show that biochar is effective for removing heavy metals from stormwater, more research should be done to better quantify predictable removal rates and inform media selection.

Takeaways: Biochar amendment has been shown to consistently reduce Cu and Zn in stormwater compared to typical bioretention media, and to effectively remove other heavy metals equally well. Batch studies including the selected biochar product should be run prior to estimating removal effectiveness for other heavy metals. Further, biochar product selection should consider CEC, and the potential for in-situ ageing that could improve biochar heavy metal removal over time (see Section 2 – Biochar Characteristics, Effects, and Stormwater Goals).

Nutrients

Nitrogen removal in media-based stormwater BMPs has predominantly been investigated using column studies. In a column study by Akpinar et al. (2023), biochar was added to two bioretention soil medias (BSM) – 1) 88% sand, 8% fines, and 4% sawdust, referred to as NC mix, and 2) 60% sand, 30% mulch, and 10% compost, referred to as DE mix. When biochar is mixed at a rate of 18% by volume into the NC mix, total nitrogen (TN) and nitrate concentrations and loadings were reduced by 65% as compared to the NC mix without biochar. 20% of the TN reduction was attributed to additional water retention in the media, and 45% attributed to nutrient reduction. The reduction of nitrate observed is expected to be due to denitrification in the micro-anoxic biochar pores, which has also been observed in other studies (Berger et al., 2019, Boehm et al., 2020, Rahman, 2020, Chen et al., 2022, Ulrich et al., 2023). An 18% biochar amendment by volume into the DE mix resulted in a decrease in TN concentration (relative to the control), but TN was still higher than the influent concentration, which has been observed in other studies where compost is included in the BSM specification (Herrera Environmental Consultants, 2015). In the same study, Akpinar et al. (2023) did not measure any significant difference in TP removal when NC mix was amended with biochar. There was an

increase in TP when the DE mix was amended with biochar as compared to the control, which is difficult to explain, though some short-term leaching of TP has been observed in studies conducted in the state of Washington, USA using certain biochar (Herrera Environmental Consultants, 2015). The extractable TP of the wood-based biochar was relatively high but since it only increased in the DE mix, the biochar is unlikely to be the source of the higher TP in the leachate. Additional research is needed to understand the mechanism causing TP to leach out. Zhang et al. (2021) found that benefits of biochar related to nitrogen reduction may not be maintained under high hydraulic loading scenarios, such as high flow filters, possibly due to a reduction in opportunities for denitrification under high flow conditions.

Afrooz and Boehm (2017) evaluated the impact of replacing the compost component in BSM with biochar. Columns were also evaluated when a saturated zone was present at the bottom of the column. Biochar amended BSM showed 20% to 27% reduction in TN, as compared to BSM with compost, which had minimal TN reduction or was itself a source of nitrogen. Columns with saturated zones reduced nitrate more than those without due to increased denitrification under anoxic conditions. Nitrate reduction also improved with biochar-amended media above the saturated zone. The same study measured no significant difference in TP removal when biochar replaced compost in a BSM, as all columns reduced TP by 18% to 22%. Plant survivability was not evaluated in this study. A separate column study by Xiong et al. (2019) used high-temperature, rice husk biochar to amend sandy BSM and observed a slightly better TN removal in the biochar-amended BSM (95%) than just sand (90%), likely driven by nitrate reduction through denitrification. However, Xiong et al. (2019) observed worse TP reduction in biochar-amended soils (60%) than just sand (90%).

A series of studies conducted in the state of Washington have investigated the potential use of biochar as a component of high-performance media blends that could replace the standard 60% sand / 40% compost blend that they typically used (Herrera Environmental Consultants, 2015, Howie and Lublinar, 2024). These studies were conducted because compost-containing BSMs were found to leach nitrogen, phosphorus, and copper, and many bioretention systems in Western Washington use underdrains due to low native soil infiltration rates. The initial study in this effort identified a regionally sourced commercial biochar as a component that could replace some of the benefits of compost without leaching TP or TN. This study identified a mix of 60% sand / 30% coconut coir / 10% biochar as the top performer, after which a new BSM specification was developed for use in regions of Western Washington draining to nutrient-impaired freshwater lakes. Follow up projects have confirmed these results but have also indicated that some biochar sources can generate short-term leaching of phosphorus, albeit at concentrations much lower than when compost was included.

In media-based stormwater BMPs, biochar amendment can improve TN reduction best when compost is not in the media mix, as compost is often a source of nitrogen (Akpinar et al., 2023, Pamuru et al, 2026). Additionally, some research has found that organic matter may reduce biochar performance by blocking pores and occupying potential adsorption sites (Beesley et al., 2011, Chen et al., 2022). However, compost provides readily available macro and micronutrients to vegetation, and co-application of biochar and compost can help mitigate nitrogen leaching from compost (Pamuru et al., 2026). Biochar amendment does not seem to have an impact on TP removal without additional biochar post-processing (cation impregnation) or other media amendment (iron filings) (Chen et al., 2022, Xiong et al., 2019), likely because TP removal is driven by particulate removal.

In green roof systems, incorporating biochar derived from hardwoods into substrate significantly reduced nutrient fluxes (Goldschmidt and Buffam, 2023). In order to support plant growth in unnatural and harsh

environments, green roof substrates often incorporate some sort of organic source. Furthermore, green roofs primarily only treat relatively clean rainwater as opposed to typical urban runoff. For these reasons, green roofs are often expected to be a source of nutrients, rather than a treatment BMP. Goldschmidt and Buffam (2023) found that phosphate concentrations decreased as the biochar concentration increased. The highest application rate (10% w/w mixed hardwood biochar) resulted in approximately 40% lower phosphate flux. Similarly, the addition of hardwood biochar decreased organic N leaching, leading to a net reduction in total N losses. However, the presence of vegetation within the planting bed had a greater impact on nitrogen flux.

Stormwater practices with biochar-amended soils (e.g., vegetated swales, filter strips, and dry pond retrofits) have not been directly studied for nutrient removal as this is not the primary rationale associated with these BMP installations. However, in a review of biochar as a soil amendment for agriculture and remediation/restoration, Blanco Canqui (2019) found that nitrogen leaching can be reduced in biochar-amended soils by 1) adsorption to biochar, 2) water and soluble nitrogen retention in macro and micro pores, 3) availability for plant uptake and denitrification, and 4) reduced erosion. Another meta-analysis determined that adding biochar to soils reduced nitrate leaching by an average of 12% compared to unamended soils, specifically by ~30% in sandy soils (Borchard et al., 2019). In a study investing the co-application of biochar and compost to sand media and soil at a rate of 4% total by weight, Pamuru et al. (2026) found that compost is essential for vegetation establishment, but the co-application of biochar will mitigate Total nitrogen leaching by 47% to 70% for amendment ratios of 3:1 and 1:1 compost to biochar, respectively. Although overall biochar removal and leaching reduction impact in agricultural soils high in nitrogen may be different, these data can likely be extrapolated to stormwater applications.

Takeaways: Biochar-amended BSMs tend to outperform sandy BSMs regarding nitrogen reduction but may reduce nutrient flux in BSMs already high in organic content (such as compost or woodchips). Biochar amendment may, though, provide a reasonable alternative to compost in flow-through BMPs to reduce the total export of nutrients. Research on the agricultural use of biochar suggests that plant health in biochar-amended soil should be strong, but plant survivability in a sand/biochar media without compost may require more resilient plant species. Biochar amendment does not improve TP removal compared to typical BSM.

Biological Contaminants

Biological contaminants include fecal indicator bacteria (FIB, typically includes fecal coliform, *E.coli*, *Enterococci*), fecal pathogens (including strains of *E.coli*, salmonella, Cryptosporidium, Giardia, etc.), and viruses. Many of the studies on the impact of biochar on biological contaminant removal have been on FIB (Abit et al., 2014, Boehm et al, 2020, Rahman, 2020, Mitchell et al., 2023). Generally, high-heat, wood-based biochar outperformed biochar from other feedstocks, increasing *E.coli* removal from the 0 to 50% figures found in initial testing, to 50 to 99% when added to a typical sand filter. This is likely due to the high hydrophobicity associated with the increased porosity and surface area (Abit et la., 2014, Suliman et al., 2017, Boehm et al., 2020, Mitchell et al.,2023). Removal efficiency was dependent on the target biological species (Suliman et al., 2017, Boehm et al., 2020), as biochar-amended soils tend to remove pathogen species better than non-pathogenic species of bacteria (Abit et al., 2014, Suliman et al., 2017). Rahman (2020) observed that *E.coli* had a higher affinity for biochar than *Enterococci* during a column study using biochar amended sand columns. In the single study on virus removal, biochar media amendment either had no or a negative impact on virus removal (Sasidharan et al., 2016). Particle size distribution also appears to have an impact on the ability of biochar to improve biological contaminant removal. Sasidhara et al. (2016) found that when fine

biochar particles (<60 µm) were removed, bacteria removal efficiencies of *E. coli* decreased from 70% to 10%, where the sand filter (control) was able to remove 30%. Similar results were observed by Mohanty at Boehm (2014). This is likely due to the increased porosity and surface area associated with finer biochar particles. It is important to note that this study used biochar produced from crop-residue feedstock, as opposed to woody feedstock, and these results may not be readily transferable, since wood-based biochar tends to have higher porosity than biochar from other feedstocks (Mohanty et al., 2018). Mohanty et al. (2014) found that biochar-amended sand reduced the remobilization of biological contaminants due to increased duration of high moisture conditions, strengthening hydrophobic bonds between microbes and biochar.

Takeaways: Biochar can improve biological contaminant removal in BMPs using sand media filtration. Biochar products should be wood-based with a low C/H ratio and hydrophobic characteristics to maximize removal. A smaller particle size distribution will likely perform better, but careful consideration should be given to necessary design infiltration rates, as smaller biochar particles reduce hydraulic conductivity (K_{SAT}) and can lead to media clogging.

Trace Organic Compounds

Trace organic compounds (TrOCs) found in stormwater can include herbicides, insecticides, motor oils, polyaromatic hydrocarbons (PAHs), polychlorinated biphenyls (PCBs), flame retardants (i.e. organophosphate flame retardants, OPFRs, and perfluoroalkyl and perfluorooctanesulfonic acid, PFAS/PFOS), and plastic manufacturing products (i.e. Bisphenol A, BPA). In a review paper on how biochar improves stormwater management, Boehm et al. (2020) found that removals of TrOCs are generally greater when stormwater is treated through biochar-enhanced BSM than when treated through sand, soil, and/or compost-amended BSM. Most studies to date on TrOCs have been using biochar to remove BPA and atrazine. Limited studies exist on biochar's impact on PFAS/PFOS and less common herbicides (i.e. simazine, oryzalin). The initial removal mechanism for TrOCs is adsorption, but biochar can also enhance long-term removal via biodegradation (Guo et al., 2020, Chen et al., 2022). Biochar can enhance TrOCs removal by increasing water retention and duration in pore spaces, increasing opportunities for sorption, and allowing time for biodegradation processes to occur (Guo et al., 2020). Dissolved Organic Matter (DOM) impacts adsorption of larger TrOCs to a higher degree than small TrOCs (Ulrich et al., 2017b). In batch studies, Ulrich et al. (2017b) found that some biochar types are more cost-effective options than activated carbon for TrOC adsorption. Further, in BMPs with high hydraulic loading and high sediment, it is unlikely that the greater sorption for certain TrOCs found in activated carbon would be utilized prior to media clogging or sorption site occlusion. These batch studies used biochar with particle sizes between 53 and 246 µm to achieve homogeneous surface properties. This particle size is smaller than typically specified in media-based stormwater BMPs, but the results still suggest that biochar would be an effective filtration media for TrOCs. Spahr et al. (2022) found dicamba, fipronil, terbutryn to be least retained in biochar-amended biofilters as compared to diuron and atenolol. Overall performance of biochar-enhanced medias with regards to TrOC removal will depend on biochar's physical characteristics, including particle size distribution, porosity and surface area, and stormwater characteristics, including DOM content and contaminant type (Ulrich et al., 2015, Mohanty et al., 2017, Pritchard et al., 2023).

Takeaways: Biochar has effectively removed TrOCs in lab-scale batch and stormwater simulated tests, suggesting that biochar would improve the removal of these contaminants compared to unamended media filters, and potentially be as effective as media filters amended with activated carbon. However, given the

variability in behavior of different TrOCs and the varying behavior of TrOCs with respect to water chemistry, it is recommended that simple batch tests are performed when targeting particular TrOCs or when numeric discharge targets are involved.

Runoff Reduction

Infiltration and Filtration Rate

Infiltration, in the context of runoff reduction, assumes that runoff and its associated pollutants are removed from the surface level water cycle. This assumption isn't always true, as the pollutants can enter groundwater (and thereby potentially impact drinking water) or seep through the subsurface to a surface waterbody. While the natural filtering process of in-situ soils can remove *some* pollutants, it may not remove *all* pollutants. In general, pollutants removed via physical filtering (e.g., metals, TP) and cation exchange have limited mobility in soils; whereas pollutants like nitrate, which are removed via anion exchange capacity, have higher mobility in soils. Infiltration is considered by most stormwater regulatory agencies as a runoff reduction and pollutant load reduction practice and is therefore considered as such in this report.

The addition of biochar to soils impacts both hydraulic conductivity and bulk density. Bulk density can be an indicator for compaction, which reduces the capacity of soils to infiltrate stormwater runoff. Biochar-amended soils tend to have lower bulk densities (Adhikari et al., 2022, Blanco-Canqui, 2017, Blanco-Canqui, 2019, Akpinar, 2023, Goldschmidt and Buffam, 2023) due to biochar's low density and high porosity. In a review of the impacts of biochar on soil properties, Blanco-Canqui (2017) found that the inclusion of biochar reduced soil bulk densities between 3% and 31%, depending on soil texture (i.e. proportion of sand, silt, clay).

Infiltration, typically measured by saturated hydraulic conductivity (K_{SAT}), is impacted by biochar amendments depending on the existing soil texture. Hydraulic conductivity can increase when biochar is applied to clay and silt soils (Edeh et al., 2020, Yan et al., 2021) as the increase in macropores improves the soil structure (Adhikari et al., 2022) and the fact that biochar is typically larger than silt and clay particles. However, Nakhli et al. (2021) found that the addition of biochar reduced the effective K_{SAT} in fine soils with high clay content. Additionally, hydraulic conductivity has been shown to decrease in sandy soils with the application of biochar (Edeh et al., 2020, Yan et al., 2021, Blanco-Canqui, 2017), likely resulting from the increased tortuosity associated with biochar-enhanced aggregation. It is notable that media-based stormwater BMPs tend to have sandy-textured soils, and that, excepting cases of high flow media filters, there may be benefits in reducing K_{SAT} to increase opportunities for pollutant removal. However, consideration of BMP design is important for determining how a biochar amendment will affect stormwater performance. Specifically, biochar used in sand-based media filters should not be significantly finer than the sand particle size that typically comprises the majority of a bioretention soil media.

BMPs such as filter strips, vegetated swales, and soil amendments are not typically considered for pollutant reduction but have the potential to improve stormwater flow and volume by slowing the flow rate of runoff and by increasing infiltration, respectively. Increasing infiltration within these BMPs will achieve greater pollutant load removal. Imhoff et al. (2020) investigated the impact of a high-temperature, pine-based biochar amendment within roadside embankments (often accounted for as filter strips leading to roadside swales) with tillage to a 30 cm depth. This study measured K_{SAT} as 1.3 to 13 times higher after two years with the biochar amendment when compared to tillage impacts alone. Akpinar (2023) found that tilled-only sites did see initial improvement compared to control sites, but only the tilled sites amended with biochar maintained

higher saturated hydraulic conductivity rates over the full study period. The greatest improvements in K_{SAT} were observed for 18% by volume biochar application rate (Akpinar, 2023, Imhoff et al., 2020).

Filtration-based BMPs, such as bioretention, biofilters, and rain gardens, are intended to achieve pollutant removal through media filtration, and are designed based, at least in part, on stormwater loading capacity. In these scenarios, higher media filtration rates (i.e., media K_{SAT}) are desirable to keep BMP footprints small, prevent ponding, and limit untreated bypass. In a mesocosm study of biochar-amended BSM, Akpinar (2023) found that after the initial BSM application, but before plants were established, there was not a statistical difference in K_{SAT} between the control BSMs (one sandy mix without compost and one sandy mix with compost) and those amended with 9% or 18% biochar by volume (2% and 4% by weight, respectively). However, after 30 weeks, K_{SAT} had significantly increased from 30 cm/hr to 48 cm/hr in the BSM amended with 4% biochar by weight and without compost. The biochar mix with compost and biochar amendment saw a slight decrease in mean K_{SAT} that was not statistically significant.

Takeaways: Biochar amendment will improve the infiltration properties of silt, and to a lesser extent, clay soils, by improving soil structure through aggregates. This is of particular interest in stormwater BMPs such as filter strips, vegetated swales, and soil amendments. However, there may be less expensive modifications for Hydraulic Soil Group (HSG) D soils, and it would be best to consult local soil experts in these regions. Media-based BMPs designed to maintain higher infiltration rates do not appear to be negatively impacted by the addition of biochar amendments up to 18% by volume, so long as biochar particle sizes are not significantly smaller than the primary media components. It is generally recommended that, to maintain infiltration rates, the biochar particle size distribution is similar to that of the primary components of the BSM.

Retention and Detention

There is an abundant amount of research showing that biochar as an agricultural soil amendment reduces runoff (Li et al., 2018) and increases water holding capacity in soils (Adhikari et al., 2022, Edeh et al., 2020, Blanco-Canqui, 2017, Kim et al., 2021), particularly in sandy soils. Biochar interacts with physical and chemical soil properties, such that biochar-amended soils show an increase in pore space and pore size diversity (Adhikari et al., 2022, Kim et al., 2021). Mesopores between 30 and 75 µm, created by soil aggregation encouraged by biochar, help retain water due to suction and capillary interactions while remaining available to plants. Micropores between 5 and 30 µm, usually intrinsic to the biochar grain, retain and hold water as well. This water is less readily available to plants, but it is still considered a component of Plant Available Water (PAW) calculations due to its presence prior to wilting. In a review of WHC for biochar-amended soils, Adhikari et al. (2023) evaluated depths up to 60 cm and found that application depths between 0 and 20 cm have the most impact on both WHC and PAW. Yi et al. (2020) developed a model based on soil and biochar characteristics to predict the changes in WHC when a soil is amended with biochar.

Biochar applications in stormwater BMPs also show an increase in water retention capacity compared to unamended soils and media. Kim et al. (2021) found that amending sandy roadside soils with 2.5% rice husk biochar processed at 400°C to a depth of 10 cm resulted in consistently higher soil water content than control soils during an extended period of drought. This was attributed to the increase in the biochar-amended soil's measured pore volume compared to the control, likely resulting from the high internal porosity of biochar. In the roadside embankments study described in the previous subsection, moisture content within the soils was approximately 45% greater in biochar-amended sites than in tilled-only sites. When calculated across the entire site, this resulted in increased runoff infiltration between 36% and 224% compared to tilled-only sites

(Imhoff et al., 2020). Akpinar et al. (2023) found that the roadside embankments amended with biochar could infiltrate between 37% and 67% more annual rainfall compared to control embankments.

Biochar is most effective at improving soil-water properties in coarse grained soils due to its potential for improved soil aggregation (Blanco-Canqui, 2017, Edeh et al., 2020). It increases the number of slowly draining pores in soils whose pores are primarily large and fast draining (Razzaghi et al., 2020). Biochar-amended sandy soils have enhanced soil structure and soil aggregation (Blanco-Canqui, 2017, Edeh et al., 2020). Islam et al. (2021) measured a 16.4% increase in soil aggregation using a high-heat wood-based biochar but measured no increase in aggregation in alkaline soils. Additionally, biochar applications in sodic soils results in reduced aggregation and the associated infiltration and water retention benefits (Interview: Hegberg). Straw-based biochar increased WHC by 18% due to macroaggregate formation (Adhikari et al., 2023). Carbon-rich biochar appears to be the most effective at forming macroaggregates (Adhikari et al., 2023). Most media-based stormwater BMPs are coarse-graded or sandy, so improved soil-water properties observed in sandy soils are very likely applicable to stormwater media mixes.

In studies on the addition of biochar in media-based stormwater applications, Akpinar (2023) found that stormwater runoff retained in the bioretention media increased from 46% to 52% when mixed with 18% biochar by volume. Based on the test site's water holding capacity and plant health, evapotranspiration (which initially accounted for 40% of water loss) increased to 50% water loss via evapotranspiration when measured in the biochar-amended media. This result suggests that the biochar amendment could increase the volume reduction potential of stormwater BMPs. In compost-containing BSMs, there was no statistical improvement in stormwater retention or ET loss over the 20-week study period when biochar was added. The improvement in the sandy BSM without compost was attributed to biochar's ability to improve macropore formation in sandy soils (Akpinar, 2023a, Akpinar, 2023b). Regarding both infiltration and runoff reduction, biochar and compost amendments to sandy BSMs function similarly by creating a soils structure that increases water retention, and supports plants and evapotranspiration (Akpinar, 2023a, Akpinar, 2023b). While no longer-term studies currently exist, it is possible that biochar could provide these benefits for the full design lifetime of stormwater BMPs whereas compost would not due to decomposition over time.

Biochar amendments to green roof media have been shown to increase porosity and the associated water retention capacity for these systems while decreasing bulk density (Beck et al., 2011, Chen et al., 2021, Laio et al., 2022). This allows for media beds to be deepened without added weight, which increases water storage potential and plant root zone depth. One study investigating green roofs also found that this increased water holding could be attributed to a reduction in TN export (Beck et al. 2011).

Takeaways: Biochar amendments can improve the hydrologic properties for a variety of stormwater BMPs. For filter strips, vegetated swales, urban soil amendment, and dry pond retrofits, adding high-temperature, wood-based biochar amendments up to 18% by volume (4% by weight) to a depth of 30 cm is expected to increase infiltration and reduce runoff volume, reduce bulk density, and increase porosity. Soil studies unrelated to stormwater biochar applications show that lower-temperature, wood-based biochar may be able to achieve similar results in these BMPs. In media-based stormwater BMPs, biochar may be a good substitute for the compost component by providing many of the same water holding benefits. BSM with larger compost fractions have been shown to leach nutrients and organic matter, whereas high-temperature wood-based biochar under similar scenarios does not (Akpinar, 2023a), biochar appears to be a reasonable alternative to compost in regions where nutrient pollution is a concern.

Plant Survivability

Plants in stormwater BMPs are important for runoff reduction (i.e. evapotranspiration, infiltration) and pollutant removal (i.e. adsorption, uptake, and as a host for bio transforming microbes). Healthy vegetation is also crucial for public acceptance of stormwater green infrastructure. However, for stormwater BMPs in harsh urban conditions, plant survival can be low, resulting in poor stormwater management performance, expensive operation and maintenance costs to replace plants, and, in some cases, a negative perception of green stormwater infrastructure. Vegetation planted along roadsides is often planted in compacted fill soils with minimal topsoil. In roadside embankments with loamy soils amended with 9% or 18% wood-based biochar by volume (2% and 4% by weight) produced at 950°C, Akpinar (2023a) measured increased vegetation cover as compared to control and till-only plots, although no significant difference was found in vegetation cover between the two biochar application rates. The author attributes this difference to the increase in plant available water measured in biochar-amended plots.

Biochar amendments also increase plant survivability in media-based stormwater BMPs. Kim et al. (2021) found a 39% increase in plant biomass under simulated drought conditions in mesocosms mimicking roadside tree boxes. These boxes were amended with low-heat wood-based biochar as compared to sandy soils in the control. The increase in plant biomass was also associated with increased soil porosity. BSMs both with and without a compost component that were amended with the 4% bw high-temperature wood-based biochar saw increased plant height and a greater number of shoots over a period of 20 weeks (Akpinar, 2023) compared to those without biochar. Although the plant height increased in both biochar-amended BSMs, the improvements were most notable in the BSM without compost. Additionally, Akpinar (2023) found that with biochar-amended BSMs, vegetation took longer to wilt during periods of extended drought conditions. The time to wilting increased from 27 to 35 days. In BSMs with and without compost, the wilting time took 57 and 52 days, respectively. Although biochar can improve plant health in compost-poor soils, Pamuru et al. (2026) found that mesocosms with biochar, compost, and co-amended sandy soils, plant viability was significantly increased when compost was part of the amendment, highlighting the importance of readily available nutrients for vegetation establishments. Goldschmidt and Buffam (2023) found that the addition of 10% wood-based biochar by volume (5% by weight) to green roof substrate saw 10% more vegetative cover after the first growing season, and recorded an additional 3% increase in vegetative cover after the second growing season than the control (no biochar) due to the access of PAW.

Takeaways: The positive impact of biochar amendment to plant vigor, health, and survivability is well documented in many applications, including the harsh environments in many urban stormwater BMPs. Biochar amendment improves soil structure by lowering bulk density (i.e. less compaction) and capturing and storing runoff and the associated nutrients in macropores and mesopores to be used by plants during drought conditions. These benefits may be more significant in the western U.S., where drought during the growing season is more common. As droughts trend more commonplace on a global scale, however, these benefits may become more broadly relevant. Although no study directly measures the cost savings of extended plant health, incorporating biochar into stormwater BMPs will save on plant replacement and/or water costs associated with these systems.

Section 4 – Concerns, Considerations, and Knowledge Gaps Around Biochar

Despite biochar's increased use in stormwater BMPs, knowledge gaps remain due to limited data or a lack of access to information from academic research. Perhaps the most prevalent concern revealed in both research and expert interviews is the variability in products, resulting in variability in performance. Additional concerns expressed are leaching of undesirable compounds and stability within the environment.

Product Variability

As illustrated throughout this document, biochar's physical and chemical properties can vary widely based on feedstock and production practices (i.e., temperature), but the resulting products are all "biochar." Although there is inherent variability in performance when relying on nature-based processes, the targeted use of biochar in stormwater BMPs allows for a reliance on quantifiable and predictable results in stormwater management applications. The typical approach to product variability in the engineering world is to develop product specifications. The existing biochar certifications create some guidance for biochar products, but these are not specific to the stormwater field. As a result, it is difficult to parse through studies when they use "certified biochar," but all employ different production processes and feedstocks. Furthermore, biochar products from similar, but not identical feedstocks and production processes, can vary in properties enough to impact study results (Interview: Erickson). This can create further confusion.

Some researchers interested in using biochar in stormwater BMPs are trying to directly link specific physical or chemical properties of biochar, such as CEC or SSA, with biochar-amended stormwater BMP performance (Fassman-Beck et al., 2025). Identifying these correlations can provide target biochar characteristic testing ranges and estimate associated pollutant reduction for biochar application in stormwater BMPs. Although, as noted in Section 2 – Biochar Characteristics, Effects, and Stormwater Goals, some biochar properties (i.e. CEC) will change over time due to ageing effects, likely improving overall BMP performance.

Other stormwater professionals recommend performing batch tests with biochar using regional stormwater conditions to select a biochar product and estimate its performance in local stormwater applications (Interview: Erickson). This option is the most desirable, as it will provide the most applicable results, but is costly for small and medium BMP applications. Therefore, general biochar specifications for use in stormwater BMPs, as developed through this project, are critical, with the option for enhanced and/or specified BMP performance if supported by local batch studies.

Stormwater Pollutant Leaching

There is concern over biochar leaching potential stormwater contaminants, which has been shown to occur for heavy metals, nutrients, and PAHs (Beesley and Dickinson, 2011, Khan et al., 2022). In most of the cases with evidenced leaching, biochar feedstock was mineral- and nutrient-rich (i.e. biosolids) or processed at low temperatures (Sia et al., 2025). BMPs experience concentrated water flow, which has the potential to leach metals, nutrients, and other potential pollutants based on water chemistry and other conditions. Although applications may exist for high-mineral biochar products in the stormwater BMPs, they will be site-specific and selected under the guidance of biochar and stormwater experts. Generally, using wood-based biochar

produced under high-heat conditions ensures that potential stormwater pollutants are not available for mobilization and leaching.

Biochar produced using high-heat pyrolysis may concentrate P and other minerals (i.e. Mg, Ca) within biochar during the reduction of mass associated combusted compounds (Cao and Harris, 2010). However, at high heat, remaining P and minerals are either contained within residual ash or they are glassed into the biochar structure and cannot be remobilized without the slow microbial breakdown of the biochar. Therefore, measurements of TP or other minerals contained within the biochar material are not an indicator of leaching potential.

The leaching potential of a biochar material can be evaluated through existing ANSI and International Standards Organization (ISO) tests. In high-heat wood-based biochar, these recommended tests will measure the potential stormwater pollutants available in the residual ash component. Studies that investigate leaching from the ash fraction of biochar show an initial flush of nutrients as the ash component is washed through the system (Blanco-Canqui, 2019, Park et al., 2015). Park et al. (2015) found that TP load associated with the initial flush plateaued to 0.1 and 0.3 mg/L for biochar produced at 300°C and 600°C, respectively after the first hour. Often, leaching tests will only capture the initial flush of pollutants and are not reflective of the biochar's long-term leaching potential. In the stormwater world, the initial flush of ash from low-ash biochar is a negligible pollutant load compared to watershed contributions, or, if present, any leaching from compost components. Therefore, it is recommended that leaching tests for biochar applications in pollutant-sensitive areas either be performed on washed biochar or incorporate a pre-flushing component to the testing procedure for the most accurate indicator of biochar leaching potential.

Biochar Stability in the Environment

One of the attractive qualities of biochar is its stable carbon structure, leading to microbial degradation resistance and weathering and, therefore, providing a long lifespan (estimated at hundreds to thousands of years) within the environment without need for replenishment (Ippolito et al., 2020). However, chemically stable biochar is structurally brittle. Ageing and weathering can be accelerated by physical forces during construction and the normal lifespan of biochar-amended BMPs, which may have performance implications.

Lifecycle in Stormwater BMPs

Common questions associated with media amendments are "how long will it last?" and "what do you do with them once they are spent?" (Kaya et al., 2022) Unlike many media amendments, biochar's chemical properties associated with pollutant reduction regenerate through ageing. As microbes slowly break down biochar structure and functional groups in soil organic matter sorb to the biochar, additional pollutant sorption sites become available for future pollutant removal. In many cases, the retention of pollutants results in the microbial breakdown (and remediation) of stormwater pollutants, which, in turn, makes previously used sites available again. In the application of high pollutant loading, there may be appropriate concern around biochar becoming spent, and these situations should be considered based on their unique site circumstances. In general stormwater applications, biochar is significantly less likely to become spent as compared to other common media additives, such as iron filings or RAC, which do not regenerate. However, further research on the fate of biochar and its sorbed pollutants would be helpful to identify any long-term BMP maintenance needs.

Stability within Stormwater BMPs

Biochar is physically very friable, which makes it susceptible to being crushed during installation and use (Spokas et al., 2014, Le et al., 2020). If biochar is crushed, its particle size distribution becomes smaller, altering its performance related to infiltration (Le et al., 2020). Once the correct biochar material is selected, this is perhaps the biggest concern associated with biochar-amended infiltration practices like high-flow media filters and bioretention (Interview: Fairbaugh). Additionally, smaller biochar particles and any sorbed stormwater pollutants are more susceptible to erosion than larger biochar particles (Spokas et al., 2014, Blanco-Canqui, 2019).

Biochar is most susceptible to fracturing during handling. These points include during transport, while mixing media, and during the process of integrating the media into the stormwater BMP. Detailed specifications and construction methods can reduce the risk of biochar fracturing; media mixing recommendations include methods which reduce particle shearing, such as using a tumbler or the "bubble-wrap" method during blending (developed during the "Scaling Up Biochar" project[3]. Additionally, ensuring the biochar and biochar-amended media meet the appropriate moisture conditions during handling allows for optimal blending without erosion (too dry) or clumping (too wet). Detailed specifications are provided within the *Biochar in Stormwater BMPs Guidebook for Stormwater Practitioners.* Compaction during and after installation should be prevented, which is typical practice for standard stormwater BMPs.

One of Many Potential Amendments

When appropriately used, biochar can improve stormwater BMP performance through pollutant removal, changes in soil hydrology, and plant survivability. However, biochar is not a silver-bullet solution, and one should practice caution when making this argument. In many cases, biochar's contribution to enhanced performance can be considered in conjunction with other media and/or soil amendments.

Biochar and Compost

Studies show that the interaction between biochar and compost is complex. Biochar has the potential to mitigate some nutrient leaching in compost (Herrera Environmental Consultants, 2015) and retain nutrients in the soil matrix longer for plant use (Goldschmidt and Buffam, 2023). Biochar and compost also provide additional improved water quality and hydrology in bioretention systems. Buates et al. (2024) applied 2% each of compost and biochar (by weight) to a pilot-scale BSM and measured improved reductions in TP and many metals. Additionally, increased levels of microbial enzymes within the media suggest enhanced microbial activity. On the other hand, organic matter has the potential to block biochar sorption sites (Beesley et al., 2011, Chen et al., 2022), which may decrease the removal of larger pollutant compounds. Biochar does not fully replace the organic component required for sustaining plants in terms of hydrologic and nutrient needs, as seen in the reduced plant health in test columns using little to no compost (Akpinar, 2023). Comprehensive studies of varying compost to biochar ratios or regional batch studies may help provide guidance on the optimal amounts of compost and biochar to include in media to achieve ideal stormwater mitigation and BMP resilience.

[3] https://www.scalingupbiochar.com/

Other Amendments

Studies repeatedly show that biochar-amended stormwater BMPs do not outperform traditional BMPs for TP removal. Studies incorporating biochar into media with additional amendments specifically targeting P removal (e.g. iron filings) show increased pollutant reductions, as compared to using a single media amendment (Xiong et al., 2019, Tian et al., 2019, Haile et al., 2026). Depending on regional stormwater goals, biochar may play a role as an ingredient in more comprehensive stormwater BMP media mixes. Further studies are needed to ensure positive interactions between biochar and other soil amendments of interest.

Conclusion

Biochar has many benefits when incorporated as an amendment into stormwater BMPs to improve the effectiveness and resilience of such practices. There is abundant research documenting biochar performance for pollutant reduction and soil improvement in stormwater BMPs. Variability in conditions, biochar type, and experimental design highlight the need for continued research and site-specific evaluation and application when adding biochar into BMPs. This technical basis report addresses this gap by summarizing general biochar performance expectations related to specific biochar characteristics in stormwater BMPs based on a literature review and interviews with stormwater and biochar experts. This report supports the recommendations for biochar use provided in the *Biochar in Stormwater BMPs Guidebook for Stormwater Practitioners* (LINK), developed with funding from the United States Endowment for Forestry and Communities by providing research and data that defends the included guidance. The guidebook provides recommendations for incorporating biochar into stormwater BMPs, including material selection, testing, and installation.

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